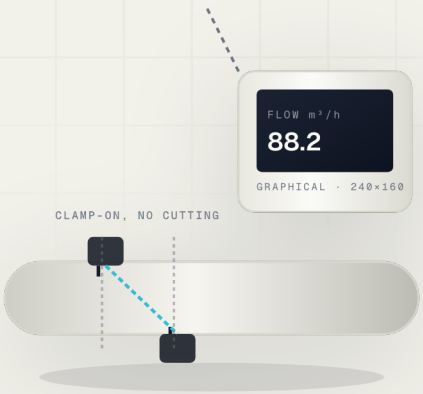
VEPOLINK
MONITORING PURITY

PROCESS INSTRUMENTATION × FLOW MEASUREMENT

NON-INTRUSIVE FLOW MEASUREMENT, NO CUTTING REQUIRED

ClampWave Flowmeter *Clamp-On Ultrasonic*

Strap-on transducers measure flow through an existing pipe wall — no cutting, no process interruption, and no wetted parts at all — with a fully graphical transmitter and three sensor versions for every duty.



12.7mm–10m
PIPE SIZE RANGE

±12 m/s
BIDIRECTIONAL VELOCITY RANGE

1 %
ACCURACY OF FLOW RATE

CLAMP-ON NO PIPE CUTTING | **3 SENSOR TYPES** ANY DUTY | **VEPOLINK** FIELD-PROVEN BUILD

0 wetted parts
SENSORS MOUNT OUTSIDE THE PIPE

±0.25%
REPEATABILITY (ISO 11631)

0.001 m/s
FLOW SENSITIVITY, RATE-INDEPENDENT

6 params
CUSTOMISABLE ON ONE SCREEN

01 OVERVIEW

Flow, measured *from outside the pipe.*

A clamp-on ultrasonic flowmeter that straps externally to an existing pipe — nothing is cut, welded or inserted into the line. Digital signal processing delivers high accuracy and strong noise immunity, holding up even with aeration or suspended solids in the liquid.

A high-speed update rate and a fully graphical display with setup wizards make installation and day-to-day operation straightforward, while a removable microSD data card and a built-in USB service port keep servicing simple.

REPORTED CHANNELS

- INSTANTANEOUS FLOW RATE
- TOTALISED VOLUME
- 4-20 MA OUTPUT
- RELAY OUTPUT
- PULSE / FREQUENCY OUTPUT
- MODBUS RTU
- TOTALISER START / STOP / RESET
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

No Cutting, No Downtime

Transducers strap onto the outside of an existing pipe — install without draining, cutting or interrupting the process.

No Pressure Drop

With no part of the sensor inside the pipe, there is no obstruction, no pressure loss and no energy loss.

Fully Graphical Display

A 240×160 graphical screen with setup wizards and up to six customisable parameters on view at once.

Easy Servicing

A removable microSD data card and a built-in USB service port make data export and firmware service straightforward.

02 MEASUREMENT PRINCIPLE

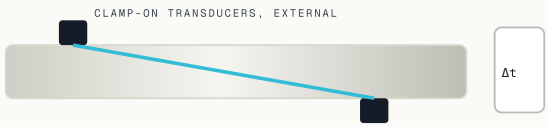
How flow is read *through the pipe wall.*

Two transducers, strapped to the outside of the pipe, send ultrasonic pulses through the pipe wall and the liquid, diagonally across the flow. The same transit-time principle as an inline sensor applies — just without touching the process at all.

TRANSIT-TIME

Through-Wall Pulse Timing

EXTERNAL TRANSDUCERS · PIPE WALL · LIQUID PATH



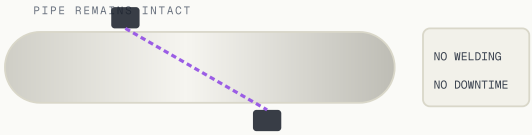
Pulses travel diagonally through the pipe wall and the liquid between two strapped-on transducers. The difference in transit time between the two directions gives flow velocity.

`Δt ∝ velocity`

INSTALLATION

Non-Intrusive by Design

NO CUTTING · NO WELDING · NO SHUTDOWN



Because the sensors never enter the pipe, the measurement adds no wetted part, no leak path and no downtime to fit.

`Pipe stays exactly as it is`

SENSOR RANGE

Three Versions, One Transmitter

WIDE-BEAM · UNIVERSAL · HIGH-TEMPERATURE



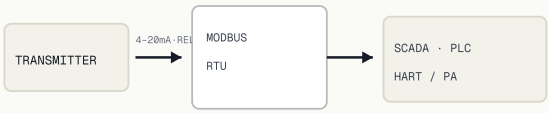
A wide-beam sensor for the highest precision, a universal sensor for general duty, and a high-temperature sensor for hot process lines all pair with the same transmitter platform.

`Match the sensor to the duty`

OUTPUT

Transmission & Reporting

4-20 MA · RELAY · PULSE · MODBUS RTU



The transmitter linearises, displays and totalises the signal, with analog, relay and pulse outputs alongside Modbus RTU, HART and PROFIBUS PA.

`Rate · total · diagnostics`

03 SENSOR SELECTION

One transmitter, *three sensor duties.*

The transmitter pairs with whichever clamp-on sensor fits the pipe, temperature and precision the application calls for — all from the same display and output platform.

SENSOR 01

Wide-beam, high precision

A wider acoustic beam for the tightest accuracy and repeatability, suited to custody-grade and process-critical duty.

SENSOR 02

Universal

General-purpose clamp-on sensor covering the broadest range of standard pipe sizes and materials.

SENSOR 03

High-temperature

Rated for hot process and utility lines where standard sensor mounting cannot be used.

PLATFORM

Shared transmitter

All three sensor types connect to the same graphical transmitter, display and output set.

NOISE IMMUNITY

Digital signal processing holds accuracy even on liquids with aeration or suspended solids — conditions that would degrade a simpler clamp-on design — with exceptional zero stability that minimises the need to set a zero point.

WIDE TURNDOWN

A ±12 m/s bidirectional range and 0.001 m/s flow sensitivity, independent of flow rate, give a wide usable turndown from one sensor on one pipe.

WHY THIS MEASUREMENT MATTERS

Adding flow measurement to a pipe that cannot be cut, drained or shut down is a recurring need across HVAC, power and process plants. Clamp-on ultrasonic measurement answers that directly, trading a small amount of accuracy headroom against an inline sensor for a fit that needs no pipework modification at all.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

Parameter	Volumetric flow, via clamp-on transducers
Working Principle	Transit-time ultrasonic, non-intrusive
Pipe Size Range	12.7 mm to 10 m
Velocity Range	±12 m/s, bidirectional
Flow Sensitivity	0.001 m/s, flow rate independent
Accuracy	1% for velocities ≥ 0.3 m/s
Repeatability	±0.25%, per ISO 11631
Working Temperature	-10 to 50 °C (transmitter)
Storage Temperature	-20 to 60 °C

SYSTEM · BUILD · POWER

Display	240 × 160, fully graphical
Channels	Single channel
Sensor Versions	Wide-beam · universal · high-temperature
Transducer Material	Stainless steel AISI 316 or similar
Pipe Material Range	Steel EN 1.0345-P235GH and most common pipe materials
Digital Inputs	2× for totaliser start/stop and reset
Output	1× 4-20 mA · 1× relay · 1× pulse/frequency
Communication	Modbus RTU
Power Supply	100-240 VAC @ 20 VA or 11.5-28.5 VDC @ 10 W
Enclosure	Polycarbonate · IP65 (NEMA 4)
Dimensions (W×H×D)	176 × 240 × 87 mm
Servicing	Removable microSD data card · built-in USB port

TRANSMITTER DIMENSIONS

INDICATIVE FRONT & SIDE · MM



05 INSTALLATION & UPKEEP

Strapped on, *running the same day.*

> Transducers strap onto the outside of the pipe using the supplied mounting rail and coupling gel — no draining, cutting or welding. Pipe outer diameter and wall material are entered into the setup wizard, and the system is ready to read **the same day.**

01 · PIPE PREP

Clean, bare metal

Clean the mounting area to bare pipe and apply the coupling gel so the ultrasonic signal transfers cleanly through the wall.

02 · STRAIGHT RUNS

Away from fittings

Mount on a straight section clear of valves, bends and pumps, per the setup wizard's recommended spacing.

03 · SENSOR SELECTION

Match the duty

Choose wide-beam, universal or high-temperature sensor per the pipe's temperature and the precision the application needs.

04 · TRANSMITTER SITING

Wall or panel mount

The graphical transmitter mounts on a wall or panel within cable reach of the clamp-on sensor pair.

05 · SETUP WIZARD

Guided configuration

Enter pipe outer diameter, wall thickness and material; the wizard calculates transducer spacing automatically.

06 · SERVICING

microSD & USB

Export logged data via the removable microSD card or the built-in USB service port without opening the enclosure.

CHOOSING WHERE TO CLAMP ON

Pick a section of pipe that runs full, is free of heavy scale or lining on the inside wall, and sits away from flow disturbances. The setup wizard's straight-run guidance keeps the acoustic path reliable once the sensors are mounted.

COMMISSIONING

Commissioning covers confirming signal strength and quality at the chosen mounting point, a zero check with the line full and static, and verifying output scaling against the plant's control or billing system — the transmitter's graphical display shows signal confidence throughout.

06 APPLICATIONS

Where it *works.*

Any pipe where flow measurement is needed but cutting into the line, draining it, or shutting the process down is not an option.

HVAC Systems	Power Engineering	Water & Wastewater
Chemical Industry		

AUDITING AND VERIFICATION

Because it needs no process interruption to install, a clamp-on meter is also the practical way to spot-check or cross-verify a permanently installed flowmeter elsewhere on the same line, or to add temporary metering during an energy or water audit.

RETROFITTING OLD PIPEWORK

Aging or hard-to-access pipe runs that were never built with a flow metering point can still be instrumented, since nothing about the existing pipe needs to change.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Transit-time ultrasonic, the same as an inline sensor, except the transducers strap onto the outside of the pipe rather than mounting inside it.

Q Does the pipe need to be cut or drained?

No. The sensors clamp externally, so installation needs no cutting, welding, draining or process shutdown.

Q What pipe sizes does it cover?

12.7 mm to 10 m, with wide-beam, universal and high-temperature sensor versions for different pipe and process conditions.

Q How accurate is it compared to an inline meter?

1% of flow rate at velocities above 0.3 m/s, with ±0.25% repeatability per ISO 11631 — close to inline performance without touching the process.

Q Does aeration or suspended solids affect the reading?

Digital signal processing gives strong noise immunity, maintaining accuracy even with aeration or suspended solids present.

Q What outputs are available?

One 4–20 mA, one relay and one pulse/frequency output from the transmitter, plus Modbus RTU communication.

Q How is it configured?

Through guided setup wizards on the fully graphical display — enter pipe diameter, wall thickness and material and the system calculates mounting spacing.

Q Can up to six parameters be shown at once?

Yes, the graphical display supports up to six customisable parameters on screen simultaneously.

Q How is data exported for servicing?

Via a removable microSD data card or the built-in USB service port, without opening the transmitter enclosure.

Q What maintenance does it need?

Minimal — periodic checks of coupling gel and signal strength at the mounting point. There is nothing inside the pipe to wear.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

FLOW, MEASURED WITHOUT TOUCHING THE PIPE

Know your flow. *No cutting required.*

Tell us the pipe size, material and the duty, and we'll match the sensor version and configure the transmitter for your line. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that sizes, specifies and commissions clamp-on flowmeters. No call centre.
+91 99538 86178 →

02 **Request a sizing & configuration**
RECOMMENDED
Send your pipe size, material and duty — we'll match the sensor version and output.
sales@vepolink.com →

03 **Book a demonstration**
See a running meter and the graphical transmitter with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM